Answer **one** question from this section in the spaces provided.

7 **(a)** Using energy considerations, distinguish between *electromotive force* and *potential difference*.

.....

.....

.....

..... [2]

(b) A cell of e.m.f. of 1.5 V and internal resistance 1.0 Ω is connected to a resistor X and resistor Y as shown in Fig. 7.1. Resistor X has resistance 2.0 Ω while resistor Y has a resistance of 6.0 Ω.

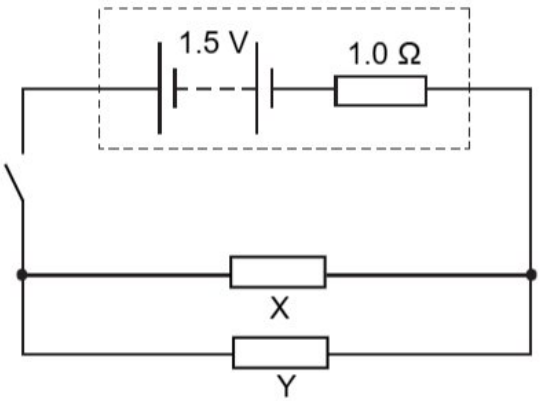


Fig. 7.1

(i) Show that the current in the cell is 0.60 A when the switch is closed. [2]

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(ii) Determine the energy dissipated in the cell when the switch is closed for 8.0 minutes.

energy dissipated = ... J [1]

(iii) Resistor Y is replaced with a component Z with similar resistance value. When the temperature increases, the resistance of component Z decreases. State and explain the change to the power dissipated in the cell when temperature increases.

.....

.....

..... [2]

(iv) The *I*-*V* characteristic of X and Z are given in Fig. 7.2.

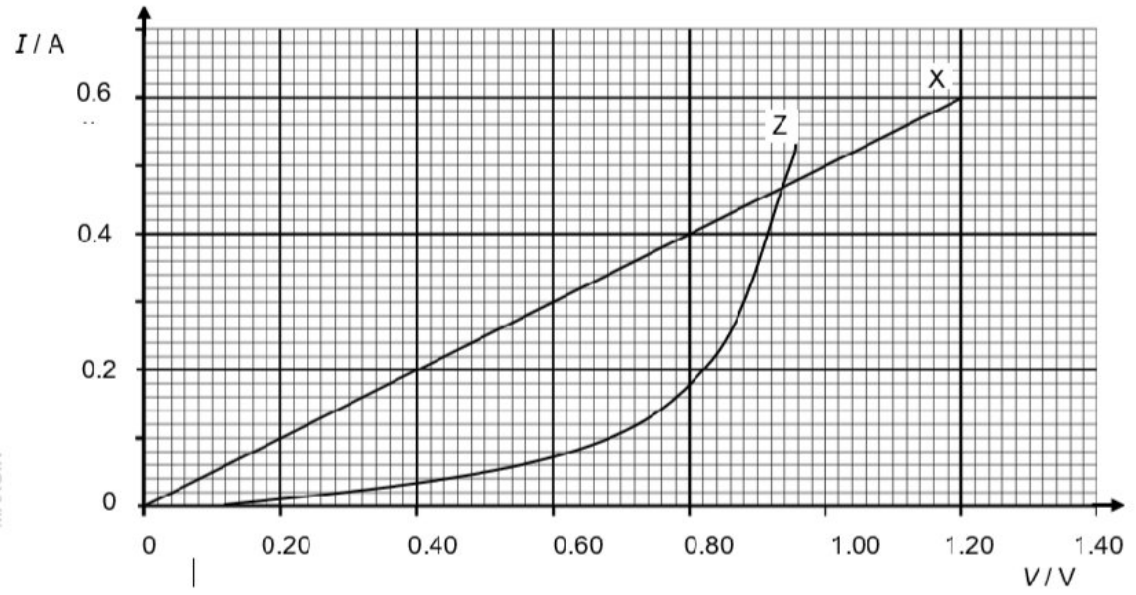


Fig. 7.2

The circuit is now reconnected such that resistor X and component Z are in series with the same cell. Using the Fig. 7.2, or otherwise, determine the potential difference across component Z.

potential difference = ... V [2]

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(v) State what the component could be.

.....

[1]

(c) A graph of the voltage input to an ideal transformer is shown in Fig. 7.3. The step-down transformer with a turn ratio of 45:1 has a mean input power of 25 W.

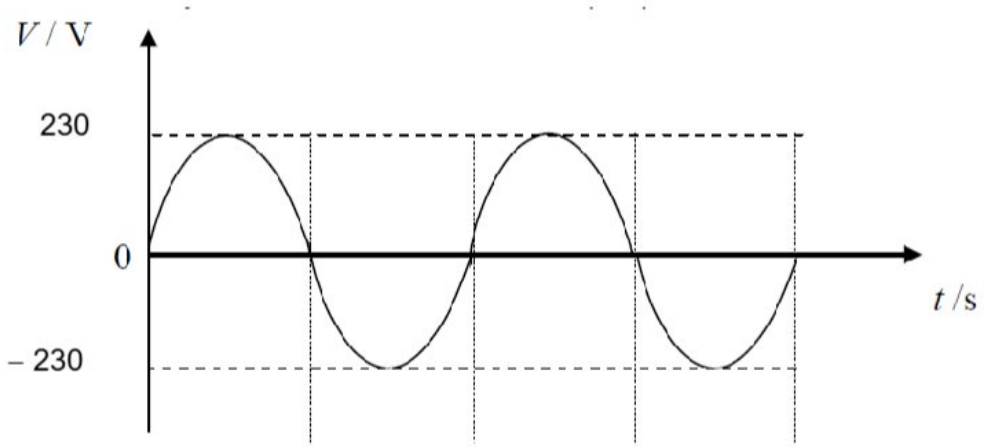


Fig. 7.3

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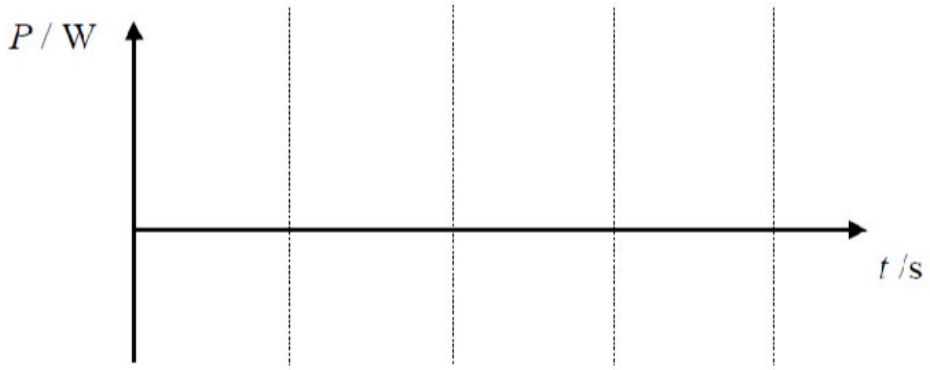


Fig. 7.4

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(i) On the *P-t* graph in Fig. 7.4, sketch the variation of power input to the transformer with time for two complete cycles. Indicate the value for the maximum input power. [2]

(ii) With reference to Fig. 7.4, explain why the mean power dissipated in the resistor is equal to half the maximum power.

.....

.....

.....

[2]

(d) Fig. 7.5 shows a simplified circuit diagram of a laptop adaptor connected to the chargeable battery of a laptop. The adapter consists of a step-down ideal transformer of turn ratio 45:1 with a single diode.

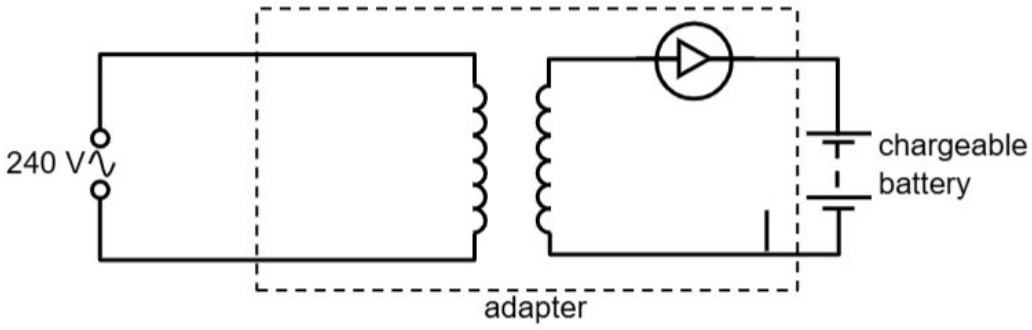


Fig. 7.5

(i) Calculate the peak voltage across the secondary coil and state one assumption made in performing the calculation.

peak voltage = ... V [2]

(ii) Explain why, during the charging process, the power delivered to the secondary coil is twice the power delivered to the chargeable battery.

.....

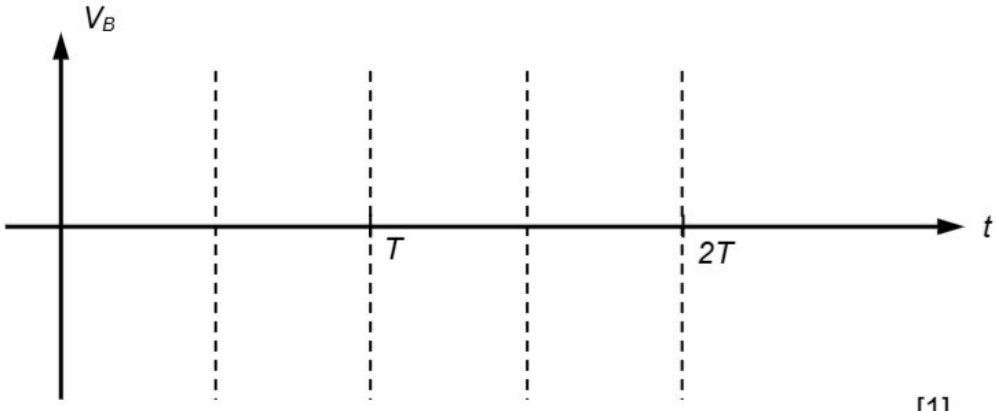
.....

..... [1]

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(iii) Sketch on Fig. 7.6 the variation of potential difference V_B across the chargeable battery with time t. It is given that T is the period of the alternating current.

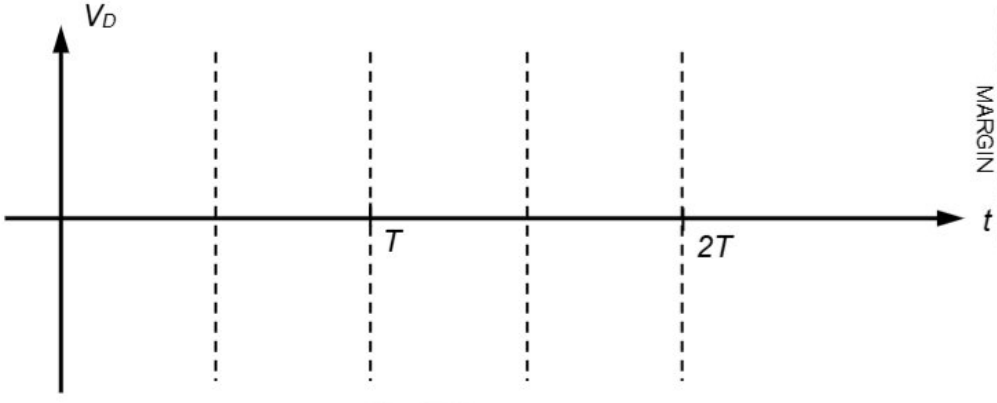


[1]

Fig. 7.6

(iv) Hence sketch on Fig. 7.7 the variation of the potential difference across the diode V_D with time t. Label appropriate values. It is given that T is the period of the alternating current.

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Fig. 7.7

[2]

[Total: 20]